Practical No. 5

Survey Report on Types of Blockchains and Their Real-World Use Cases

Aim:
To study the different types of blockchain networks and their real-world applications, advantages, and limitations.

1. Introduction

A **blockchain** is a distributed digital ledger that records transactions in a secure, transparent, and tamper-resistant manner. Instead of storing data on a single central server, blockchain stores copies of the ledger across multiple computers called **nodes**.

Blockchain technology is mainly used for **cryptocurrency**, but it is also used in banking, healthcare, supply chain, voting, digital identity, and many other areas.

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2. Types of Blockchains

There are mainly **four types of blockchains**:

1. Public Blockchain

A public blockchain is open to everyone. Anyone can join the network, verify transactions, and participate according to the network's rules.

Features:

- Open and decentralized
- High transparency
- Anyone can participate
- Generally uses consensus mechanisms such as Proof of Work or Proof of Stake

Examples:

- Bitcoin
- Ethereum

Real-time use cases:

- Cryptocurrency transactions
- Decentralized applications (DApps)
- Smart contracts
- DeFi applications

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2. Private Blockchain

A private blockchain is controlled by a single organization. Only authorized users can access and participate in the network.

Features:

- Permission-based access

- Faster transactions
- Better privacy
- Controlled by an organization

Real-time use cases:

- Banking systems
- Internal business records
- Enterprise data management
- Supply-chain management

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3. Consortium Blockchain

A consortium blockchain is controlled by **a group of organizations** rather than a single organization.

Features:

- Semi-decentralized
- Only selected organizations can participate
- Faster and more efficient than many public blockchains
- Suitable for collaboration between organizations

Real-time use cases:

- Banking and financial institutions
- Insurance companies
- Supply-chain networks
- Inter-organizational data sharing

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4. Hybrid Blockchain

A hybrid blockchain combines features of **public and private blockchains**. Some information can be made public while sensitive information is restricted to authorized users.

Features:

- Combination of public and private access
- Better privacy and transparency
- Flexible
- Controlled access to sensitive information

Real-time use cases:

- Healthcare records
- Government services
- Real-estate records
- Supply-chain tracking

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3. Comparison of Blockchain Types

Feature	Public	Private	Consortium	Hybrid
Control	Decentralized	One organization	Multiple organizations	Mixed
Access	Anyone	Authorized users	Selected members	Partly public
Transparency	High	Low/controlled	Controlled	Selective
Speed	Comparatively lower	High	High	High
Example	Bitcoin, Ethereum	Enterprise blockchain	Banking networks	Enterprise/government systems

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4. Real-Time Applications of Blockchain

1. Cryptocurrency

Blockchain is the foundation of cryptocurrencies such as **Bitcoin and Ethereum**. It records transactions without requiring a traditional central authority.

2. Banking and Finance

Blockchain can be used for:

- Faster international payments
- Transaction settlement
- Fraud reduction
- Digital assets

3. Supply Chain Management

Companies can record the movement of products from manufacturer to customer. This improves **traceability and transparency**.

Example:

Manufacturer
↓
Distributor
↓
Retailer
↓

Customer
Each stage can record information on the blockchain.

4. Healthcare

Blockchain can help maintain secure and controlled access to medical records and improve sharing of information between authorized healthcare organizations.

5. Digital Identity

Blockchain can be used to create verifiable digital identities, reducing dependence on physical documents.

6. Voting

Blockchain-based voting systems can potentially provide:

- Transparent records
- Tamper-resistant storage
- Easier auditing

However, secure implementation and privacy remain important challenges.

7. Smart Contracts

A smart contract is a program stored on a blockchain that automatically executes predefined rules.

Example:

Condition satisfied
↓
Smart Contract
↓
Automatic execution
Ethereum is widely used for smart contracts and decentralized applications.

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5. Advantages

- Decentralization
- Transparency
- Data integrity
- Tamper resistance
- Reduced dependence on intermediaries
- Traceability
- Automation through smart contracts

6. Limitations

- Scalability problems in some networks
- Transaction fees

- Energy consumption for certain consensus mechanisms
- Regulatory challenges
- Privacy concerns
- Technical complexity

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7. Conclusion

Blockchain is a distributed ledger technology that provides secure and tamper-resistant data storage. The four major types are **Public, Private, Consortium, and Hybrid blockchain**. Each type is suitable for different requirements. Blockchain has real-world applications in **cryptocurrency, banking, supply chains, healthcare, digital identity, voting, and smart contracts**.

Result:
The different types of blockchain and their real-time applications were studied successfully.